New Zealand Army
NZ1 – Trentham HF Station
MINUTE

23 Oct 26

OIC Ex NS 26 (Lt D.J.G. Crowe), CFSCE Kingston

For information
NZ3 (Linton)
NZ4 (Burnham)

POST ACTIVITY REPORT: EXERCISE NOBLE SKYWAVE 26 – NZ1

References

A. CFSCE 3352 Ex NOBLE SKYWAVE 2026 Exercise Instructions, 24 Jun 26.

B. Ex NS 26 Annex A – CEOI (incl. Appx 2 Frequency List, Appx 3 Call Schedule), 28 Sep 26.

C. Ex NOBLE SKYWAVE Fragmentary Instructions 002, 5 Oct 26.

D. Ex NS 26 Web App scoreboard, https://nobleskywave.ca/scoreboard, as at 22 1948Z Oct 26.

E. Noble Skywave Station Map (NZ1), https://benhaddley.github.io/noble-skywave-fieldbook/, 22 Oct 26.

F. Noble Skywave – App Workflow Showcase (simulated data), 23 Oct 26.

Note: this report supersedes the interim report of 22 Oct 26 (results as at 22 0919Z). It was completed after End Ex (22 1600Z Oct 26). Scores are as published on the Web App scoreboard at 22 1948Z Oct (Ref D) and remain subject to confirmation by exercise control.

Introduction

1. Ex NOBLE SKYWAVE 26 (Ex NS 26) is an annual international HF skywave competition run by the Canadian Forces School of Communications and Electronics (CFSCE), with the Net Control Station (NCS) at CFB Kingston, ON. The competition ran from 20 1200Z to 22 1600Z Oct 26 (end time amended by Ref C). 219 teams from Canada, the US, Peru, Germany, Australia, Romania, Iraq and other nations registered scoring activity.

2. NZ1 was a New Zealand Army team at Trentham, Upper Hutt, operating in the SSB Team Contact group and the 1–150 W power category. Two other NZ teams took part: NZ3 (Linton) and NZ4 (Burnham). NZ1 also ran an automated radio desk trial, using software to coordinate checks over the exercise Slack, make voice calls, and record and transcribe returns.

Executive summary

3. NZ1 was heard in South America and Japan from New Zealand at about 40 W, worked PT1 (Portugal) two-way by voice over 18,114 km, and finished with 110.5 points. Most of these points (102.5) came in Stage 4, where distance is rewarded and NZ1 placed 80th of 180. NZ1 did not score in Stage 1: the NCS did not acknowledge either of NZ1's calls. The dominant limitation was receive: across 304 receive checks with 92 stations, the automated desk recovered partial copy from only five, and for the last four hours of Stage 4 a stalled receive audio stream stopped most checks. NZ1's transmissions were reported heard at readability 3/3 by PE2 and PE40 and heard by

JP1, so the circuit was asymmetric. The automated desk worked within its safety controls throughout and is worth developing, but receive capability and the transmit antenna need to be fixed before NS 27.

Objectives

4. NZ1's objectives for the exercise were:

   a. to confirm NZDF HF skywave interoperability with international teams from New Zealand;

   b. to establish voice contact with the NCS and score in each competition stage;

   c. to exercise the FLEX-8600 station and its antennas at Trentham under competition conditions; and

   d. to trial automated coordination, calling, recording and transcription within a strict operator approval and transmit-safety regime.

Summary of Activities

5. The exercise was conducted in three phases (Ref A), with the competition in four stages (Ref B).

   a. **Phase 1 –** Preparation (to 20 Oct). Registration (deadline 16 Oct), CEOI build into the station channel plan, receiver reference recordings at 10 and 15 MHz on both antennas, a short authorised voice transmit test on 13.953 MHz (20 Oct; 6.7 W peak, SWR 1.56:1) and NZ team coordination with NZ3 and NZ4. NZ1 also published a searchable map of registered stations, with distance and bearing tools, for all teams (Ref E); it was refreshed from the registration map on 22 Oct.

   b. **Phase 2 –** Competition (20–22 Oct):

      (1) Stage 1, Establish the Net (20 1400Z). With no New Zealand slot in the calling schedule, NZ1 called in the Sydney slot. Two calls were made at 20 2216Z and 2217Z (MEDIUM power, 44 W peak, SWR 1.3:1 or better). No return was heard and the NCS did not acknowledge. Earlier the same day, peer checks with AU8, AU17 and CA26 on 13.953 and 11.538 MHz were not heard. NZ3 achieved two-way contact with the NCS on 24.700 MHz.

      (2) Stage 2, Free Play (21 0200–1200Z). Calls to PE20, PE40 and CA89. Two of these were cut off by the SWR/power interlock (see para 11).

      (3) Stage 3, Team Contact Challenge (21 1200Z–22 0200Z). Calls to PE28 and AU4 within the SSB group; no two-way contact confirmed.

      (4) Stage 4, Back to the Future (22 0200–1400Z). The bulk of NZ1 activity: 25 desk transmissions (one interrupted) to Peru, Canada, the US, Germany, Romania, the UK, Japan, South Korea and Australia between 4.8 and 20.3 MHz, with the call budget renewed at 0947Z. Reports received from PE2 (3x3), PE40 (3/3), PE9 (2x2), JP1 (heard) and UY1 (1/1). At 0837Z NZ1 worked PT1 (Portugal) two-way on 9.9425 MHz (18,114 km, 6 points). From 1010Z the receive audio stream stalled repeatedly, so returns to the last eight calls were not captured (see para 10).

   c. **Phase 3 –** Post-ex drills (22 Oct onward): final results recorded (para 6); an illustrated walkthrough of the station software, using simulated data, produced for briefing (Ref F); AAR points to OIC Ex NS 26 NLT 20 Nov 26.

Results

6. Final scores as at 22 1948Z Oct, after End Ex (Ref D).

Stage	NZ1 points	NZ1 rank	NZ3	NZ4	AU17 (ref)
2.1 Establish the Net	0	Not placed	1.00	–	0.50
2.2 Free Play	7.00	154 of 174	67.50	20.00	85.00
2.3 Team Contact (SSB group)	1.00	115 of 118	15.00	6.00	14.00 (3G group)
2.4 Back to the Future	102.50	80 of 180	326.50	8.50	229.50
Total	**110.50**		**410.00**	**34.50**	**329.00**

Resources

7. The following equipment was used:

   a. FlexRadio FLEX-8600 worked from a Maestro control head, with automatic antenna tuner;

   b. ANT1 wideband dipole (all receive; transmit from 22 0701Z) and ANT2 vertical (transmit until 22 0628Z), as recorded by the station software;

   c. control laptop running the station console: radio control, Slack coordination, text-to-speech voice calls, recording and Whisper transcription;

   d. public KiwiSDR receivers for independent monitoring and, from 22 Oct, a HackRF One integrated as a second, receive-only local receiver; and

   e. the NZ1 station map (Ref E) for distance, bearing and path planning.

8. Activity totals: 36 peer calls, two of them interrupted, and 2 NCS calls (on 22 Oct the desk used all 24 calls in its first budget and 8 of a further 25 authorised at 0947Z); 304 receive checks with 92 stations; 8,147 recordings (27.0 hours of audio) and 6,866 transcripts; 29 Web App reporting drafts.

Summary of Key Observations

9. The following observations are those considered key for NZ1. The transmit call record is at Annex A.

10. **Observation: Receive performance.** NZ1's main limit was receive, not transmit. Of 304 receive checks, 169 heard nothing and only five stations (UY1, CZ2, CA74, US131, PE9) produced partial copy. Meanwhile PE2 and PE40 copied NZ1 at 3/3 and JP1 heard NZ1 on 9.941 MHz. The Trentham receive noise floor sat around S4–S5 on the dipole with no signal present. Where signals were strong, receive worked: the operator copied PT1 loud and clear on 9.9425 MHz at 22 0834Z. From 22 1010Z the receive audio stream stalled repeatedly, and 21 of the 26 checks after that time stopped with 'receive audio stalled'. The desk correctly declined to report those stations as not heard, but the fault was not cleared before Stage 4 closed, which was overnight in New Zealand.

    a. **Recommendation:** conduct a noise survey at Trentham and trial a dedicated low-noise receive antenna separated from the transmit antenna, using the HackRF as a second receiver.

    b. **Recommendation:** pre-arrange remote receive (e.g. KiwiSDR or a partner station) as an approved second ear for logging, if the rules allow.

    c. **Recommendation:** alert the operator as soon as the receive audio stream stalls, and hold further calls until receive is restored or the operator accepts transmit-only working.

11. **Observation: Transmit antenna and power.** ANT2 (vertical) tripped the SWR interlock on 15.733 MHz (3.84:1, aborting the AU17 call), 20.9 MHz (14.7:1) and 6.95

MHz (8.8:1). Separately, HIGH power overshot to about 115 W on 23 MHz and tripped the 110 W watchdog. NZ1 therefore ran at MEDIUM (36–47 W peak) for the rest of the exercise. On 22 Oct ANT2 matched at 1.2–1.3:1 and carried ten calls to 0628Z; from 0701Z transmit moved to ANT1, which matched at 1.2–1.5:1. The interlocks worked as designed but cost three calls.

a. **Recommendation:** sweep and repair ANT2 and its feed before NS 27; characterise tuner matches across the CEOI frequency list in advance.

b. **Recommendation:** calibrate the HIGH power setting per band, so the station can use its full 1–150 W category allowance.

12. **Observation: NCS calling schedule.** Ref B gives no New Zealand slot in the NCS calling schedule. NZ1 called in the Sydney slot, which is a poor fit for propagation from Kingston to Wellington at that hour. Neither call was acknowledged, costing up to 3 points. NZ3 later made two-way contact with the NCS on 24.700 MHz.

    a. **Recommendation:** request a dedicated NZ/Pacific slot for NS 27, timed against VOACAP for the Kingston–NZ path, and seek a pre-exercise test with the NCS.

13. **Observation: Coordination and logging.** Most contacts were arranged in the exercise Slack. Teams usually responded within minutes, but many offers expired or were cancelled (107 of 304 receive checks) as teams moved frequency or went off shift. Scoring requires both teams to log the same contact within 5 minutes, so a contact the peer did not log earned nothing. NZ1 raised 29 Web App reporting drafts and submitted 26: 8 were confirmed on submission (5 accepted, 3 matching a report the peer had already made) and 18 were awaiting the peer's report.

    a. **Recommendation:** keep using the chat for skeds, but agree a frequency, a time and a logging plan in a single message, and log on the Web App as soon as the peer confirms.

14. **Observation: Automated desk.** The desk worked within its call budgets (24 calls, then a further 25 from 22 0947Z; 3 per station), with operator approval for transmissions, EMCON and SWR/key-time interlocks. No unsafe transmission occurred. Issues found and fixed during the exercise were: the software answering chat before the operator; a coordination panel that showed stations as 'Calling' when they had only been invited; and two processes overlapping, which interrupted a DE10 call.

    a. **Recommendation:** develop the desk further before NS 27; keep the approval gate and RF interlocks; complete the interlock testing before any unattended operation.

    b. **Recommendation:** limit desk automation to coordination, logging and recording, with a human operator on the microphone for marginal circuits.

15. **Observation: National teams.** NZ3 outscored NZ1 in every stage and was the strongest NZ team, finishing Stage 4 12th of 180 with 326.5 points (410 in total). The three NZ teams shared frequencies and results in a group chat, which helped; NZ1 passed PT1's frequency to NZ3 straight after its contact.

    a. **Recommendation:** share station and antenna lessons with NZ3 and run a joint NZ pre-exercise net for NS 27.

16. **Observation: Exercise documentation.** Ref C moved mission end to 22 1600Z, superseding Ref A (2000Z). Annex A's Back to the Future list gives 20.2685 MHz, which does not appear in Appendix 2 (20.2695 MHz). Voice dial offsets differ between the Annex A prose (1.5 kHz) and Appendix 2 (1.4 kHz).

    a. **Recommendation:** include these discrepancies in AAR points to the OIC.

17. **Observation: Records and documentation.** The station software kept records of desk calls, receive checks, recordings, transcripts and Web App drafts, from which this report was compiled. An illustrated walkthrough of the workflow, using simulated data, is at Ref F.

    a. **Recommendation:** retain the software, logs and recordings for NS 27, and use the walkthrough to brief incoming operators before the next exercise.

Conclusions

18. Ex NS 26 was a valuable activity for NZ1. From New Zealand at about 40 W, NZ1 was heard in Peru and Japan, worked Portugal two-way, scored 110.5 points across three of the four stages, and showed that an automated desk can run safely within tight transmit controls.

19. NZ1 did not score in Stage 1, and poor receive limited two-way contacts. Receive capability (including monitoring of the receive audio chain), the ANT2 vertical and the NCS slot for New Zealand are the priorities for NS 27.

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Annex:
A. NZ1 Transmit Call Record, Ex NS 26.

Distribution:
OIC Ex NS 26, CFSCE
NZ3 – Linton
NZ4 – Burnham
File

ANNEX A TO
NZ1 PAR EX NS 26
DATED 23 OCT 26

NZ1 TRANSMIT CALL RECORD – EX NS 26

All calls USB voice from the FLEX-8600. Frequencies are dials in MHz. Times are Zulu. 'TX-only' means the peer heard NZ1 but NZ1 did not copy the return. From 22 1010Z the receive audio stream stalled, so returns to the last eight calls were not captured. Contacts worked outside the desk, such as PT1 at 22 0837Z, are not listed.

DTG (Oct)	Station	Dial (MHz)	Peak W	Result
20 1955	AU8	13.9531	–	Failed – nil heard both ways
20 2007	AU8	11.5381	–	Failed – nil heard both ways
20 2050	AU17	13.9531	–	Failed – "no joy"; nil heard by NZ1
20 2056	CA26	13.9531	–	No report from peer
20 2137	AU17	15.7331	–	Aborted – ANT2 SWR 3.84:1
20 2216	NCS	15.733 (Sydney slot)	44.4	No acknowledgement
20 2217	NCS	15.733 (Sydney slot)	42.6	No acknowledgement
21 0926	PE20	15.7331	36.8	Failed – peer calling, nil copy
21 1050	PE40	20.9001	–	Aborted – SWR/power interlock
21 1112	CA89	6.9501	–	Aborted – SWR/power interlock
22 0105	PE28	13.5841	43.6	No report
22 0143	AU4	14.9611	35.9	Unresolved – nothing heard
22 0206	PE28	13.5841	40.8	Unresolved – "can't hear"
22 0242	US131	15.7331	38.9	Unresolved – "no dice"
22 0356	PE2	14.9611	39.4	TX-only – report 3x3
22 0357	CA79	4.8091	43.2	No report
22 0401	US59	11.5381	39.5	No report
22 0407	CA79	4.8091	40.6	No report
22 0537	AU17	14.8411	37.8	No report
22 0628	PE28	14.9611	38.9	No report
22 0701	PE9	20.3001	45.8	Unresolved – peer "copy 2x2"
22 0720	PE40	9.9411	43.2	Failed – nil copy both ways
22 0753	DE10	13.9531	42.6	Unresolved – first attempt interrupted
22 0813	PE40	7.3101	45.9	TX-only – report 3/3 "very low"
22 0815	GB8	10.5611	42.4	Failed – nothing heard
22 0817	PE41	5.8321	43.2	No report
22 0818	RO7	14.8411	36.9	No report
22 0824	PE9	7.3101	44.0	No report
22 0843	JP1	9.9411	41.5	TX-only – heard, with JP amateur QRM
22 1014	US49	7.3101	47.4	No report
22 1026	CA89	7.3101	–	Interrupted – operator claim changed
22 1104	UY1	11.5381	40.2	Unresolved – peer "rogg 1/1"
22 1213	US47	11.5381	39.8	No report
22 1217	KR1	14.3731	40.1	No report
22 1249	US52	11.5381	39.2	No report
22 1250	US52	11.5381	42.6	No report
22 1320	CA82	7.7001	43.8	Failed – "nothing heard"

Receive checks (peer transmitting, NZ1 listening): 304 checks with 92 stations – 169 not heard, 107 expired or cancelled, 6 partial copy (UY1, CZ2, CA74, US131, PE9), 22 stopped by a fault (21 stalled receive audio from 22 1010Z, 1 Slack delivery).